Contribution

Camille : My contribution focused on Task 2 of the unsupervised learning lab. With guidance, I replaced the commented example with executable code to create a K-Means model with four clusters and fit it to the dataset. I added commands to retrieve the cluster labels and center coordinates. I also prepared a scatter plot that colors the data points by cluster and marks the centers with large red X symbols. After running the cell, I encountered a NameError stating that KMeans was not defined. I shared the error for troubleshooting and identified that the earlier setup cells needed to run before continuing.

Rigoberto :

For Lab 10, my individual contribution was completing Question 1 in the Knowledge Check section of the unsupervised learning notebook. My responsibility was to explain the difference between supervised and unsupervised learning. I explained that supervised learning uses labeled data with input features (X) and known target outputs (y), while unsupervised learning uses input features (X) without predefined answers to discover patterns and relationships in the data.I completed my assigned question and focused on making sure my explanation clearly described how the two learning methods work. I wanted to show that I understood how supervised learning uses known answers to train a model, while unsupervised learning allows the model to find patterns without being given those answers.This question also helped me connect the concepts to the other activities in the lab. For example, K-Means clustering is an unsupervised learning technique because it groups similar data points without needing labeled answers. PCA is another example because it reduces the number of dimensions in a dataset while keeping important information.My contribution was focused on the knowledge check and conceptual understanding rather than the coding tasks completed by my other group members. I completed my assigned portion of the notebook and made sure my answer connected to the overall topic of unsupervised learning. Overall, I cotributed by completing Question 1 and demonstrating my understanding of the two main learning approaches. This helped me better understand why different machine learning methods are used depending on the type of data available and the problem being solved.

Min:

For this lab, I was responsible for the data setup, Task 1 (the Elbow Method), and Knowledge Check Q2. Before we started, I prepared a shared notebook for the group. The file had a formatting problem that kept it running, so I fixed it without changing any of the content. I also assign the roles for our team members. For my part, I ran the setup cell to generate the synthetic customer data. Then I completed Task 1 by fitting K-Means for k values from 1 to 10, saving the inertia for each one, and plotting the elbow curve. Inertia dropped from about 19,780 at k=1 to 362 at k=4 and then leveled off, which confirmed k=4 as the best number of clusters for Task 2. I wrote the answer to Q2, explaining why we don’t just pick the largest k.

Chris:

I was assigned to complete Task 3 (PCA) and Knowledge Check Question 3 for the group lab. For Task 3 I used the Iris data set and reduced the 4 features to 2 principal components. I calculated the explained variance in the data and then made a scatter plot of the transformed data. For Question 3 I described how PCA preserves most of the information in a data set while reducing its dimensionality. Lastly, I focused on learning to do PCA for the unsupervised learning tasks in this lab.